

Certified analytical bounds for hypersonic reachable sets

A two-page summary of a preliminary PhD research proposal

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The question. A vehicle entering an atmosphere at hypersonic speed can be steered, within limits, and neither the air nor its own aerodynamics is known better than within tolerances. What can it be driven to? The states it can reach under *every* admissible control and every admissible model form its reachable set, which a safety analysis has to enclose; a landing footprint is its shadow on the ground. This proposal is about an *outer* bound on it that is proved and not sampled, and that is a formula: explicit in the vehicle and the flight corridor, to be evaluated again, not solved for again, when either changes. The idea is that the thin atmosphere supplies one: in the limit a flight is a slow motion punctuated by brief passes through dense air, and the bound sought is the reachable set of that much smaller limit system, thickened by an explicit margin. The vehicle is a parameter of the bound, not its subject.

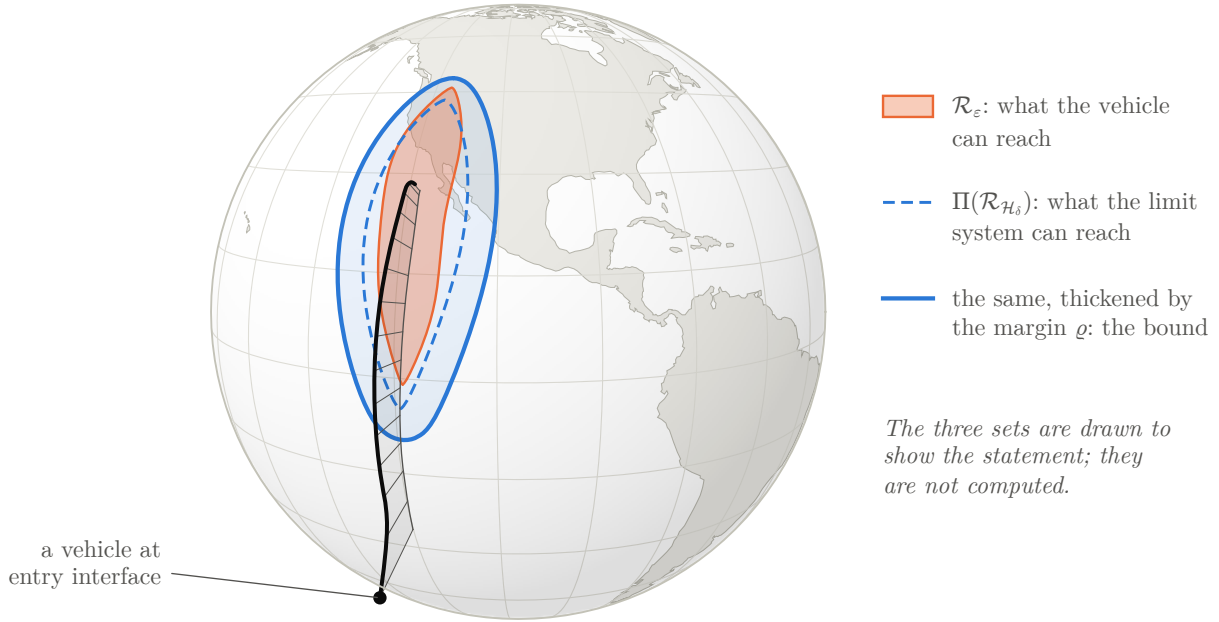


Figure 1: The statement aimed at, drawn on the ground. The trajectory is an example, a lunar-return skip entry simulated from a published entry state [1], with altitude drawn twelve times too large and a post each minute. The three sets are an illustration and are not computed: what the vehicle can reach, what the limit system can reach, and the second thickened by a margin, which is the bound. The first may leave the second, but by no more than the margin.

The gap. One outer bound is free, and it shows what is missing. Drag dissipates energy at the rate $P = DV$ per unit mass, so a path of length R costs at least $RD_{\min}$, and with $\Delta\mathcal{E}$ the energy there is to lose, $R \leq \Delta\mathcal{E}/D_{\min}$. It holds for every control because it uses nothing about the control, and it is useless for the same reason: it charges the whole flight at the thinnest air the corridor allows, so that across a skip it says nothing. The tools that use more do not reach the question. A Hamilton–Jacobi equation on a grid stops near five states [2], and a rigid body has thirteen; a flowpipe [3] encloses one vehicle, one initial set and one horizon, and tends to widen along weakly damped motion; and guidance is validated by sampling [4].

What the free bound ignores. It ignores *when* the energy is lost. The ratio of the atmosphere’s scale height to the planet’s radius, $\varepsilon = H_s/R_e \approx 10^{-3}$, is a small parameter, and as it vanishes a trajectory comes apart into three motions. Above the atmosphere it follows an arc under gravity and loses nothing. On the equilibrium glide, where lift balances weight less centrifugal relief, it loses energy slowly. And in a *pass* it dips into dense air, turns, and sheds a fixed share of its energy in a vanishing time. A pass is also where a control decides where the vehicle goes, and it is what the standard relaxation loses. In the space of occupation measures the dynamics are a linear constraint and an outer bound is certified by a dual pair of functions [5]; but an occupation measure weighs each interval of time by its length, and in the limit a pass has none.

The proposal is to keep the account in energy instead. Along every trajectory, for every control and every selection of an interval-enclosed vector field, the energy in the rotating frame satisfies $\dot{\mathcal{E}} = -P \leq 0$, so the dissipation has total mass at most $\Delta\mathcal{E}$, uniformly in ε . On the rescaled time $\tau = t/\sqrt{\varepsilon}$ its normalized density is a sequence of

probability densities on the line, and Lions' concentration-compactness lemma [6] applies as stated; the reading to be argued is that vanishing is the glide, compactness a pass, and dichotomy passes that separate. Along a subsequence the dissipation converges to a measure $p dt + \sum_j a_j \delta_{t_j} + \pi_c$, with at most $\Delta\mathcal{E}/\delta$ atoms above δ . What has to be proved first is what the atoms carry: that the velocity converges to a function of bounded variation that jumps only at them, and that the limit of a trajectory satisfies a Liouville equation with a jump term they carry.

The theorem aimed at. The atoms are to be the passes, and the limit a hybrid system $\mathcal{H}_\delta$ in a few slow coordinates. It *flows* along gravity arcs and along the glide, and it *jumps*, at most $K_\delta \leq \Delta\mathcal{E}/\delta$ times, through a set-valued *pass map*: skeleton and bubbles, in the language of profile decompositions [7]. The statement aimed at is one at fixed ε , uniform over the admissible family:

$$\mathcal{R}_\varepsilon(T) \subseteq \Pi(\mathcal{R}_{\mathcal{H}_\delta}(T)) \oplus \mathbb{B}(\varrho(\varepsilon, \delta)),$$

the reachable set of the full system inside that of the limit system, read back in the full coordinates and thickened by a ball of explicit radius, as the figure draws it. Compactness gives structure and no constants; in the wave-equation theory the analogous uniform bound is obtained by contradiction, with a constant its authors say they cannot describe [7]. Here the constants are to come from direct estimates, in closed form in the corridor bounds and the interval widths.

The obstacle. With z the altitude above the glide in scale heights and $G = g - V^2/r$ the acceleration that lift balances, the vertical motion is, to leading order,

$$\ddot{z} = \frac{G}{H_s} (c(t) e^{-z} - 1), \quad \frac{d}{dt} \left[\frac{1}{2} \dot{z}^2 + \frac{G}{H_s} (e^{-z} + z) \right] = \frac{G}{H_s} (c - 1) e^{-z} \dot{z},$$

where $c(t)$ is the control, the vertical lift against a reference. The glide is the equilibrium, the phugoid its small oscillation, and a skip its large one. The frequency grows like $\varepsilon^{-1/2}$ and the damping rate does not depend on ε , so the damping ratio is $O(\sqrt{\varepsilon})$: the glide is not a normally hyperbolic slow manifold [8], and no boundary-layer estimate brings a trajectory onto it. And the control pumps or damps the oscillation at will, by the identity on the right. The amplitude is therefore a slow, controlled coordinate of the limit system, and what is needed is an averaging theorem *with constants*, uniform over a set-valued field, and valid for *every* trajectory, including those that pass through resonance: the attitude frequency grows with the square root of the dynamic pressure and sweeps through the others on every pass. The available results give orders, not bounds, and through a resonance they hold outside a set of small measure [9], which an outer bound cannot discard. This is the critical path, and it is the part of the program for which I am looking for an advisor.

What is open. All of it: this is a proposal, and it claims no result. Three statements carry it. The limit: that the jump term is carried on the graph of the pass map, and that no singular continuous part survives. Composition: a factorization through pass maps and flows with an explicit matching remainder, uniform over the interval field, where the classical matched expansions treat one trajectory [10]. And averaging through resonance, with constants, which is the critical path and is to be attacked first.

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